

Azlan Ahmad

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Education

University of Western Ontario

Expected May 2027

Bachelor of Science in Data Science & Software Engineering — GPA: 3.9 / 4.0

London, ON

- **Awards & Certifications:** Dean's List, AWS Cloud Practitioner (CLF-C02), Microsoft Azure Fundamentals (AZ-900)
- **Relevant Coursework:** Machine Learning, Statistical Learning, Data Structures & Algorithms, Database Systems, Linear Algebra, Probability & Statistics, Software Engineering

Experience

Data Science Intern

Sep 2025 – Present

Electrical Safety Authority

Mississauga, ON

- Cut data processing time by **65%** via **Python and SQL ETL pipelines**, ingesting **millions** of inspection records
- Deployed **XGBoost** and **LightGBM** models to production, flagging high-risk inspections **weeks in advance**
- Reduced manual review workload by **40%** via **Isolation Forest** and **DBSCAN** anomaly detection pipelines
- Built automated **PySpark** data quality monitors maintaining **>99% integrity** across 3 cross-source datasets

Data Science Intern

Sep 2024 – Apr 2025

Bruce Power

Tiverton, ON

- Surfaced **\$20K+ in cost savings** for leadership by auditing **Azure** resource usage with custom **SQL** queries
- Shipped a **RAG-based** knowledge retrieval tool using **LangChain** and **LlamaIndex**, adopted by **3 internal teams**
- Ran **A/B tests** (n=5K+, p<0.05) with **Pandas** and **SciPy**, driving a workflow change that cut processing delays
- Forecast task completion rates with **scikit-learn** models, improving **KPI prediction accuracy by 22%**

Data Engineering & BI Intern

May 2024 – Aug 2024

Dartmouth College

Hanover, NH

- Reduced **ETL execution time by 30%** via parallelized **Apache Spark** jobs, processing **10M+ academic records**
- Led cloud migration to **GCP BigQuery** and **Dataflow**, cutting infrastructure costs by **50%** for the data team
- Architected an **AWS Redshift** data warehouse serving **real-time BI dashboards** for institutional stakeholders

Machine Learning Engineer Intern

Jan 2024 – Apr 2024

University of Toronto

Toronto, ON

- Achieved **92% accuracy** in sleep stage classification via **CNN**, outperforming SVM and RF baselines by **15%**
- Cut model training time from **8 hours to 2 hours** using **CUDA** and **PyTorch Lightning** GPU acceleration
- Presented model comparison findings to the research team, informing pipeline selection for production deployment

Data Science Intern

May 2023 – Apr 2024

University of Western Ontario

London, ON

- Trained a **GAN** in **TensorFlow** and **PyTorch** to generate synthetic MRI images at **85% clinical accuracy**
- Reduced query latency by **40%** by re-indexing and tuning a multi-terabyte **SQL/NoSQL** research database
- Preprocessed **1.2 TB** of medical data with **NumPy/Pandas**, removing **15%** duplicates and **20K+** mislabels

Technical Skills

Languages: Python, SQL, R, C, C++, Java, JavaScript, TypeScript

ML / Data: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, MLFlow, PySpark, LangChain

Tools & Platforms: Apache Spark, Git, PostgreSQL, MySQL, Snowflake, Tableau, Power BI, Jupyter, Docker

Cloud: AWS (Redshift, S3, Lambda), Azure (AKS, Data Factory), GCP (BigQuery, Dataflow)

Methods: Supervised & Unsupervised Learning, NLP, Deep Learning, A/B Testing, Statistical Inference, ETL/ELT, RAG